Hyeong Jun Lee

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RESEARCH INTERESTS

- **Patient-specific computational hemodynamics:** image-based coronary CFD (incl. lattice Boltzmann method) and fractional flow reserve (FFR) assessment
- **Physics-informed machine learning:** neural operators and physics-informed surrogates for real-time cardiovascular flow prediction
- **Image-based cardiovascular physiology:** CT-FFR and diffusion-based 4D coronary CT synthesis for dynamic FFR
- **Wearable hemodynamic diagnostics:** PPG-based estimation of non-Newtonian blood viscosity, blood pressure and fluid status

EDUCATION

Yonsei University — Seoul, Republic of Korea
Ph.D. in Mechanical Engineering (Integrated M.S.–Ph.D. Program) — *Mar. 2020 – Aug. 2026*
- Dissertation: "Patient-Specific Computational Hemodynamics: Flow Field Analysis for Cardiovascular Assessment"
- Advisor: Prof. Joon Sang Lee

Pusan National University — Busan, Republic of Korea
B.Eng. in Mechanical Engineering — *Mar. 2016 – Feb. 2020*

RESEARCH EXPERIENCE

Yonsei University — Seoul, Republic of Korea
Postdoctoral Researcher, Multiscale Fluid Dynamics Lab (PI: Prof. Joon Sang Lee) — *Sep. 2026 – Present*
Appointment: Technical Research Personnel (alternative military service, through Aug. 2027), Engineering Research Institute, Yonsei University
Concurrent: Project Researcher, Yonsei University Health System – 4D CT-FFR project (PI: Prof. Jung-Sun Kim, Department of Internal Medicine, College of Medicine)
- Physics-informed neural operators for real-time, patient-specific prediction of coronary flow fields (journal manuscript in preparation).
- Wearable PPG biomarkers: deep-learning ensemble that predicts fluid-bolus needs from smartwatch PPG spectrograms [M1]; blood pressure, viscosity and hydration estimation.
- Dynamic 4D CT-FFR and 4D flow-based coronary risk assessment in government-funded projects (KHIDI, NRF).

Graduate Research Assistant, Multiscale Fluid Dynamics Lab (Advisor: Prof. Joon Sang Lee) — *Mar. 2020 – Aug. 2026*
- Coronary CT-FFR: built image-based, patient-specific CFD pipelines (incl. lattice Boltzmann method) and a gray-zone FFR prediction algorithm combining hemodynamic features, synthetic vessel models and biometric data [J4].
- Physics-informed neural operators: developed a PINO surrogate (PointNet++ branch, Fourier-feature trunk, Navier–Stokes residual regularization) that predicts 3D coronary pressure and velocity fields on unseen geometries in seconds [C1].
- Dynamic 4D CT-FFR: contributed to a diffusion deformable model that synthesizes full-cycle 4D coronary CT from two CT phases for time-resolved FFR analysis [J1, C2].
- Wearable hemodynamics: designed a physics-constrained CNN-LSTM with a Carreau–Yasuda prior to estimate patient-specific non-Newtonian blood viscosity from smartwatch PPG [J2].
- Cardiac device hemodynamics: CFD-based optimization of a tricuspid regurgitation membrane device [J3].

E8 (E8IGHT Co., Ltd.) — Seoul, Republic of Korea
Research Intern — *Jan. 2021 – Jun. 2021*
- Performed lattice Boltzmann method (LBM)-based computational fluid dynamics simulations.

PUBLICATIONS

*† first author; * corresponding author*

Journal Articles (SCIE)

[J1] T.H. Han†, Y.W. Kim, **H.J. Lee**, J.S. Kim, S.G. Lee, D.H. Yang, H.M. Oh, D.S. Kim, S.Y. Shin, S. Song, J.S. Lee*, "Synthesis of coronary 4D CT image by denoising diffusion probabilistic model," *Computer Methods and Programs in Biomedicine*, vol. 282, 109382, 2026. doi:10.1016/j.cmpb.2026.109382

[J2] **H.J. Lee**†, Y.W. Kim, S.Y. Shin, S.L. Lee, C.H. Kim, K.S. Chung*, J.S. Lee*, "A Physics-Integrated Deep Learning Approach for Patient-Specific Non-Newtonian Blood Viscosity Assessment using PPG," *Computer Methods and Programs in Biomedicine*, vol. 265, 108740, 2025. doi:10.1016/j.cmpb.2025.108740

[J3] Y.W. Kim†, **H.J. Lee**, S.J. Jung, J.H. Kim*, J.S. Lee*, "Optimization of tricuspid membrane mechanism for effectiveness and leaflet longevity through hemodynamic analysis," *Engineering Applications of Computational Fluid Mechanics*, vol. 16, no. 1, pp. 1587–1600, 2022. doi:10.1080/19942060.2022.2104929

[J4] **H.J. Lee**†, Y.W. Kim†, J.H. Kim, Y.J. Lee, J.S. Moon, P. Jeong, J.H. Jeong, J.S. Kim*, J.S. Lee*, "Optimization of FFR prediction algorithm for gray zone by hemodynamic features with synthetic model and biometric data," *Computer Methods and Programs in Biomedicine*, vol. 220, 106827, 2022. doi:10.1016/j.cmpb.2022.106827

Manuscript Under Revision

[M1] **H.J. Lee**†, J. Myung, J.Y. Hong*, J.S. Lee*, "Multi-Model Deep-Learning Ensemble for Predicting 24-Hour Fluid Bolus Administration from Continuous Photoplethysmography Spectrogram Analysis," *Engineering Applications of Artificial Intelligence*, first revision under review.

Peer-Reviewed Conference Proceedings

[C1] **H.J. Lee**†, T.H. Han, J.S. Kim, S.G. Lee, J.S. Lee*, "Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators," *Proc. ASME 2026 Design of Medical Devices Conference (DMD2026)*, Minneapolis, MN, USA, V001T02A010, 2026. doi:10.1115/DMD2026-1091

[C2] T.H. Han†, Y.W. Kim, **H.J. Lee**, J.S. Kim, S.G. Lee, D.H. Yang, H.M. Oh, J.S. Lee*, "Deriving Dynamic Fractional Flow Reserve From 4D Coronary CT Synthesized by a Diffusion Model," *Proc. ASME 2026 Design of Medical Devices Conference (DMD2026)*, Minneapolis, MN, USA, V001T02A011, 2026. doi:10.1115/DMD2026-1092

CONFERENCE PRESENTATIONS

International

[I1] **H.J. Lee** et al., "Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators," ASME Design of Medical Devices Conference (DMD2026), Minneapolis, MN, USA, 2026 (poster).

[I2] T.H. Han et al. (incl. **H.J. Lee**), "Dynamic Fractional Flow Reserve from 4D Coronary CT Synthesized by a Diffusion Model," ASME Design of Medical Devices Conference (DMD2026), Minneapolis, MN, USA, 2026 (poster).

[I3] **H.J. Lee** et al., "Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators," 18th Asian Congress of Fluid Mechanics (ACFM), Seoul, Korea, 2025 (oral).

[I4] **H.J. Lee** et al., "AI-based Hemorheology Prediction with Patient-Specific Biometric Boundary Conditions," 26th International Congress of Theoretical and Applied Mechanics (ICTAM), Daegu, Korea, 2024 (short oral & poster).

[I5] **H.J. Lee**, J.S. Lee, "Optimization of Artificial Intelligence Algorithms for FFR Prediction in Gray Zone," 10th International Conference of Manufacturing Technology Engineers (ICMTE), Gyeonggi-do, Korea, 2022 (oral).

[I6] **H.J. Lee** et al., "Estimating CFD-based CT FFR using lattice Boltzmann method – 3D geometry auto segmentation and novel patient-specific computation," ESCHM-ISCH-ISB 2021, Fukuoka, Japan, 2021 (oral).

Domestic (Korea)

[D1] **H.J. Lee** et al., "A Physics-Integrated Deep Learning Approach for Patient-Specific Non-Newtonian Blood Viscosity Assessment using PPG," Biomedical Engineering Society for Circulation (BESCO) Summer Meeting, Gwangmyeong, 2025 (poster).

[D2] T.H. Han et al. (incl. **H.J. Lee**), "Time-resolved Coronary Flow Visualization from Single-phase CT," BESCO Summer Meeting, Gwangmyeong, 2025 (poster).

[D3] D.Y. Lee et al. (incl. **H.J. Lee**), "Arterial Blood Pressure and Blood Viscosity Estimation from AI-based PPG Analysis," BESCO Winter Meeting, Seongnam, 2024 (poster).

[D4] S.W. Kwon et al. (incl. **H.J. Lee**), "FFR Prediction with Quantum Machine Learning: Accurate and Non-invasive Diagnosis with Limited Features," BESCO Winter Meeting, Seongnam, 2024 (poster).

[D5] D.Y. Lee et al. (incl. **H.J. Lee**), "Blood viscosity estimation from wearable device using AI-based PPG analysis," BESCO Summer Meeting, Daejeon, 2024 (poster).

[D6] S.E. Kim et al. (incl. **H.J. Lee**), "Carved membrane design for tricuspid valve regurgitation treatment," BESCO Summer Meeting, Daejeon, 2024 (poster).

[D7] **H.J. Lee**, J.S. Lee, "Unlocking Predictive Health Outcomes with Biometric Data," Korean Society of Mechanical Engineers (KSME) Conference, Songdo, 2023 (oral).

[D8] **H.J. Lee** et al., "Modeling Coronary Artery Hemodynamics: Exploring DCNN Surrogate Models in Preliminary Research," BESCO Summer Meeting, Daegu, 2023 (oral).

[D9] **H.J. Lee** et al., "Artificial Intelligence Algorithms for FFR Prediction in Gray Zone by Single-view Angiography," BESCO Winter Meeting, Seoul, 2022 (short oral & poster; Best Paper Award nominee).

[D10] J.H. Kim et al. (incl. **H.J. Lee**), "Autonomous Cardiovascular Hemodynamics Analysis using Patient-Specific CFD Simulation with Unsupervised Learning-based Coronary Artery Segmentation," BESCO Summer Meeting, Seoul, 2021 (oral).

PATENTS

16 applications in total (12 Korea, 2 US, 2 PCT); 2 registered.

Registered

[P1] J.S. Lee, M.Y. Yoon, S.C. Ko, **H.J. Lee**, "AI-based noninvasive urodynamics test method and apparatus," Korean Patent No. 10-2902500, registered Dec. 2025 (Appl. No. 10-2022-0085583).

[P2] J.S. Lee, **H.J. Lee**, Y.W. Kim, "Optimization system and method of AI algorithm for prediction coronary artery lesions based on FFR," Korean Patent No. 10-2882193, registered Nov. 2025 (Appl. No. 10-2022-0030019).

Pending

[P3] J.S. Lee, J.S. Kim, **H.J. Lee**, S.G. Lee, "Patient-specific coronary flow prediction method based on physics-informed neural operators," KR Appl. No. 10-2026-0003225.

[P4] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, "Deep learning-based multiphase coronary CT interpolation system and method," US Appl. No. 19/397,163.

[P5] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, "Deep learning-based multiphase coronary CT interpolation system and method," KR Appl. No. 10-2025-0149607.

[P6] J.S. Lee, **H.J. Lee**, "Device and method for predicting viscosity using AI in a viscosity prediction system," PCT/KR2025/002462.

[P7] J.S. Lee, **H.J. Lee**, "Device and method for predicting viscosity using AI in a viscosity prediction system," KR Appl. No. 10-2025-0021284.

[P8] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, "Deep learning-based multiphase coronary CT interpolation method," KR Appl. No. 10-2024-0168733.

[P9] J.S. Lee, **H.J. Lee**, "Method for predicting blood glucose and diabetes using PPG signals and biometric information," KR Appl. No. 10-2024-0147541.

[P10] J.S. Lee, W.R. Choi, **H.J. Lee**, "Wearable device for monitoring glaucoma suspect and method for monitoring glaucoma suspect," PCT/KR2024/012365.

[P11] J.S. Lee, **H.J. Lee**, "Non-Newtonian fluid viscosity modeling of patient blood using wearable device-based PPG and biometric information," KR Appl. No. 10-2024-0026709.

[P12] J.S. Lee, **H.J. Lee**, "Systolic and diastolic viscosity prediction algorithm using wearable device-based PPG and biometric information," KR Appl. No. 10-2024-0028201.

[P13] J.S. Lee, **H.J. Lee**, "Glucose and diabetes prediction algorithm using wearable device-based PPG and biometric information," KR Appl. No. 10-2023-0144347.

[P14] J.S. Lee, W.R. Choi, **H.J. Lee**, "Wearable device for monitoring glaucoma suspect and method for monitoring glaucoma suspect," KR Appl. No. 10-2023-0118582.

[P15] J.S. Lee, **H.J. Lee**, "Glaucoma diagnosis method and system based on contactless biosignals," KR Appl. No. 10-2023-0078883.

[P16] J.S. Lee, **H.J. Lee**, "Optimization system and method of AI algorithm for prediction coronary artery lesions based on FFR," US Appl. No. 17/820,819.

RESEARCH PROJECTS (PARTICIPATING RESEARCHER)

[R1] "Real-Time Precision Medicine Platform Based on Smart Hemodynamic Indices," National Research Foundation of Korea (NRF), Leading Research Center Program (NRF-2022R1A5A1022977), Jun. 2022 – Feb. 2029.

[R2] "4D Coronary Flow Visualization from CT in Real Time (4D CT-FFR)," Ministry of Health and Welfare / KHIDI, Clinical-Needs Translational Research (RS-2024-00406717), Apr. 2024 – Dec. 2026.

[R3] "4D Flow-Based Diagnostic Device for Dynamic Risk Assessment of Anomalous Origin of the Coronary Artery in Children and Adolescents," Ministry of Health and Welfare / KHIDI, Pediatric Disease R&D Program (RS-2025-02263580), Apr. 2025 – Dec. 2028.

[R4] "Patient-Specific Coronary Artery Disease Risk Analysis Software," Ministry of SMEs and Startups, Startup Growth Technology Development Program (TIPS) (RS-2024-00441761), Jul. 2024 – Jun. 2026.

[R5] "Patient-Specific Implantable Device for Tricuspid Regurgitation Using 3D Heart Printing and 4D-CFD Simulation," Ministry of Health and Welfare / Korea Medical Device Development Fund (KMDF_PR_20200901_0104), Sep. 2020 – Dec. 2025.

[R6] "Variable Negative-Pressure Tunnel for Respiratory Infection Control," Ministry of Health and Welfare / KHIDI, Infectious Disease Medical Safety Technology Program (HG22C0001), Apr. 2022 – Dec. 2024.

[R7] "High-Speed Point-of-Care FFR Simulator for Cardiovascular Diagnosis and Treatment," NRF Bio & Medical Technology Development Program (2017M3A9E907337122), Feb. 2020 – May 2022.

AWARDS AND HONORS

Merit Academic Paper Award, School of Mechanical Engineering, Yonsei University (for [J2]) 2026
Best Paper Award, Biomedical Engineering Society for Circulation (BESCO) 2022
Merit Academic Paper Award, Yonsei University 2022
Best Paper Award, School of Mechanical Engineering, Yonsei University 2022

Fellowships and Scholarships

BK21 FOUR Graduate Research Scholarship (7 semesters), Ministry of Education / NRF 2022 – 2025
BK21 PLUS Scholarship, Ministry of Education / NRF 2020 – 2021
Academic Research Fellowship (ARF), Yonsei University 2021

TEACHING EXPERIENCE

Teaching Assistant, School of Mechanical Engineering, Yonsei University Mar. 2020 – Dec. 2020

SKILLS

Computational modeling: image-based, patient-specific coronary CFD; lattice Boltzmann method; hemodynamic indices (FFR, wall shear stress); non-Newtonian blood rheology

Machine learning: physics-informed neural networks and neural operators (DeepONet/PINO); surrogate modeling; diffusion models; CNN/LSTM; ensemble learning

Biomedical data: coronary CT angiography (incl. 4D CT), invasive FFR, smartwatch PPG signals; multi-institutional clinical collaborations

Languages: Korean (native), English (professional working proficiency)

REFERENCES

Available upon request.